

Relational Database Engines on Quantum Platforms: Concepts, Algorithms, and Implementations

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Abstract

Recent advances in quantum computing have led to early-stage platforms with over 1,000 qubits, alongside roadmaps targeting fault-tolerant, industry-scale systems within this decade. To harness the immense potential of these emerging systems, it is essential to explore the feasibility of hosting relational database engines on quantum platforms. Encouragingly, several early studies have already investigated the applicability of quantum computing to core DBMS components such as multi-query optimization, join-order optimization and index-configuration selection offering initial insights into key challenges and practical implementation considerations.

In this tutorial, we provide an in-depth exploration of how quantum computing can be harnessed to advance database technologies. We begin by introducing the foundational principles of quantum computation and then outline the distinct technical challenges that emerge when operating on quantum hardware. The discussion proceeds to quantum-driven optimization methods, applicable to various DBMS components such as query optimization and physical schema design. Subsequently, we examine quantum-based query execution for core relational operators, emphasizing the architectural strategies proposed to work around the probabilistic computational framework. We conclude by outlining open research problems that must be solved to realize practical quantum databases.

In the concluding segment, participants will engage in a comprehensive demonstration, followed by a practical session where they will design quantum database algorithms and deploy them on both quantum simulators and real quantum hardware.

CCS Concepts

• **Information systems** → **Autonomous database administration**; • **Hardware** → **Quantum technologies**.

Keywords

Quantum Databases, DBMS Index Advisor, Quantum Computing

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1 Overview

Advances in our understanding of quantum mechanics have, over recent decades, transformed quantum computing from a theoretical curiosity into a viable computational framework. By harnessing principles like *superposition*, *interference*, *entanglement*, *tunnelling* and *irreversible measurement*, quantum computers can perform information processing tasks far exceeding the reach of classical machines. Landmark developments in the 1990s, including Shor's factoring algorithm [39] and Grover's search algorithm [13] established the potential of quantum computation for exponential and quadratic speedups, respectively. In the past decade, hybrid variational approaches such as QAOA [10] and VQE [31] have revitalized research by coupling quantum circuits with classical optimizers to solve complex optimization problems. However, tailoring these algorithms to specific classical problem domains remains non-trivial and typically necessitates customized design adaptations.

Leading technology companies, including IBM and Google, have made significant investments in quantum computing and already developed systems with limited but promising capabilities, ranging from 100 to 1000 qubits [11, 29]. Moreover, quantum computers capable of solving industry-grade problems are no longer a distant dream. Recent breakthroughs point to a promising future that transcends classical limits [24]. The consensus expectation is that *within this decade*, fault-tolerant quantum computers capable of executing a quantum circuit with 100 million gates will be commercially available for solving complex industrial problems [11].

Building on this momentum, several recent major industry announcements have further underscored the accelerating progress toward practical *quantum advantage*. For instance, recently, IBM announced a major collaboration with four National Quantum Innovation Centers to advance quantum-centric supercomputing by integrating quantum processors with classical high-performance computing and developing scalable quantum networks for scientific applications [30]. Around the same time, Google reported a landmark milestone, demonstrating a quantum algorithm called *Quantum Echoes* that performed a physics simulation 13K times faster than the world's fastest classical supercomputer [33]. These efforts, along with several parallel initiatives, underscore the growing industrial focus on quantum computing and its potential to soon transform scientific and enterprise computation [18].

Quantum Databases

In recent years, the database research community has increasingly explored the feasibility of hosting relational database management systems (RDBMS), the foundational engine of enterprise data processing, on quantum computing platforms. This intersection promises to be a powerful catalyst for addressing the demands of Big

Compute workloads. Several vision papers have underscored the potential of quantum computing to accelerate core database operations [7, 14, 32, 42, 46, 47]. Complementing these conceptual works, empirical studies have demonstrated tangible performance improvements for various RDBMS components, including multi-query optimization [9, 35, 40], join-order optimization [26, 28, 36, 44], transaction scheduling [5, 6, 12], and index configuration [4, 22, 41]. Other efforts have focused on quantum-assisted estimation of query cardinality, cost, and execution time [43], as well as direct SQL query execution on quantum architectures [8, 15, 20, 21, 23, 45].

As quantum computing advances from theoretical constructs to deployable systems [11, 29], this research direction is gaining significant traction within the data management community. The goal of this tutorial is to cover these emerging efforts in quantum databases and emphasize their transformative potential for the next generation of data processing systems.

2 Tutorial Contents

In this tutorial, we begin with an overview of the fundamental principles underlying quantum computing. We then explore several key quantum algorithms, including Grover's Search and the Quantum Approximate Optimization Algorithm (QAOA). We then focus on quantum databases and review key research proposals in two complementary dimensions – *query optimization* and *query execution*, and highlight their open technical challenges. We conclude with a detailed demonstration and a hands-on exercise on constructing quantum database algorithms and deploying them on both quantum simulators (Qiskit AER [16]) and real quantum hardware (Qiskit Runtime Primitives [17]).

Overall, the tutorial is structured into **six stages**, detailed below, covering the research spectrum from concepts to algorithms to implementations.

2.1 Stage I: Quantum Computing Fundamentals

In this segment, we review the fundamental building blocks of quantum computing, beginning with the notion of the *qubit*, the quantum counterpart of the classical bit. We then discuss the principles of superposition and entanglement, core quantum phenomena that enable parallelism and non-local correlations, respectively. Next, we extend the discussion to the construction and manipulation of multi-qubit systems. Finally, we present an overview of standard quantum gates, such as the Pauli-X, Hadamard, and CNOT, which together constitute the basic components of quantum circuits.

We then walk through a collection of popular quantum algorithms. We begin with Grover's Search, which achieves a quadratic speedup over classical approaches for unstructured search problems. This is followed by a discussion of the Quantum Approximate Optimization Algorithm (QAOA), a hybrid variational technique tailored for solving combinatorial optimization tasks.

Finally, we examine the two primary categories of quantum hardware platforms: *quantum annealing* and *quantum circuit* models. We compare their respective strengths and limitations, offering insights into their applicability across different classes of quantum algorithms and use cases.

2.2 Stage II: Quantum Computing Challenges

In this segment, we examine the key technical challenges that arise when deploying relational database systems on quantum computing platforms. A visual overview of these challenges is provided in Figure 1, encompassing topics such as data loading and probabilistic results. Each of these aspects poses distinct obstacles to effective database management and calls for novel, quantum-aware design strategies. Some representative challenges are discussed below.

2.2.1 Quantum Data Loading. A key challenge in quantum database research concerns the loading of classical data into quantum states. At first glance, this issue may appear to have been resolved, given several successful demonstrations of quantum computing applied to RDBMS tasks [22, 36, 40]. However, each of these efforts employs a distinct data encoding strategy. For instance, in [40], query workload instances are represented as weights on and between qubits, allowing the optimal configuration minimizing the input formula to be determined via adiabatic quantum annealing. In contrast, [36] addresses the join-ordering problem through a sequence of transformations—first formulating it as a mixed integer linear programming (MILP) model, then converting it into a binary integer linear programming (BILP) form, and finally mapping it to a Quadratic Unconstrained Binary Optimization (QUBO) structure for quantum execution, as summarized in Figure 2. Further, in [22], the index selection problem is encoded within the relative phase of a quantum superposition state. These examples collectively indicate that existing approaches rely on task-specific encodings, underscoring an open research challenge: the design of a unified, general-purpose quantum data loading framework.

2.2.2 Non-deterministic Results. Quantum computations are inherently *probabilistic*, whereas DBMS workloads generally demand



Figure 1: Research Challenges in Quantum Databases

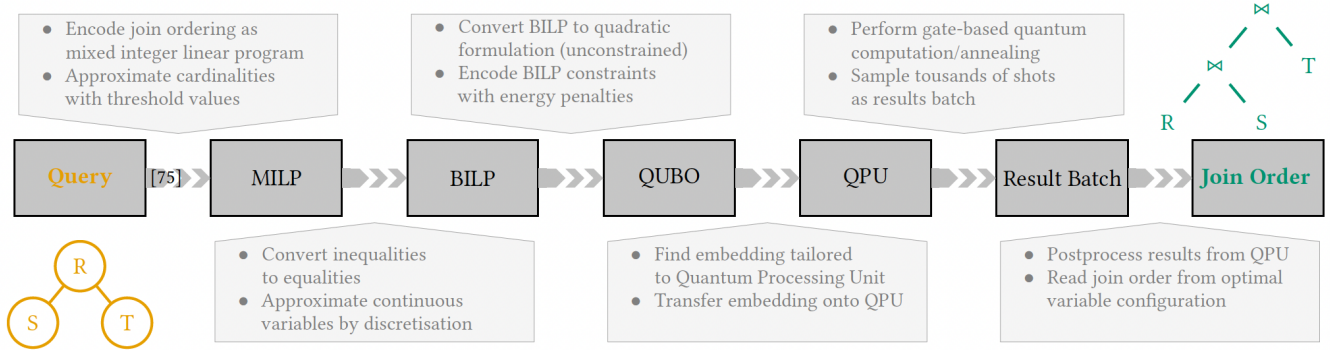


Figure 2: Quantum Computing Pipeline for Relational Join Ordering [36]

deterministic and repeatable results. Consequently, the outputs generated from quantum executions may exhibit:

- (1) **False Positives (Incorrect Results):** Quantum algorithms may incorrectly classify a wrong solution as correct. This stems from the pronounced susceptibility of quantum systems to noise and decoherence, which may cause qubits to lose their quantum coherence and collapses into erroneous states, resulting in false positives within the output.
- (2) **False Duplicates (Repeat Productions of Correct Results):** A quantum algorithm may produce the correct result multiple times. While this might initially seem beneficial, it can cause inconsistencies in subsequent stages of query processing, particularly when determining unique outputs or computing aggregates. This behavior arises from the intrinsic uncertainty and probabilistic nature of quantum states.
- (3) **False Negatives (Missing Results):** The quantum algorithm may fail to retrieve some correct solutions. For instance, in Grover's Search, the final measurement step randomly selects one of the qualifying states with equal probability. This stochastic selection can cause certain valid solutions to be repeatedly omitted, as the same state may collapse and be measured multiple times, leading to missing results.

2.3 Stage III: Quantum Query Optimization

A promising paradigm for quantum optimization is *quantum annealing*, which operates by adiabatically evolving a quantum system from an initial ground state to the ground state of a target *Hamiltonian*. The Hamiltonian is a mathematical operator representing the total energy of the system; in the context of optimization, its structure is designed such that its lowest-energy configuration corresponds to the optimal (or near-optimal) solution of the problem. Through this gradual evolution, the system naturally converges toward the state encoding the desired solution. Quantum annealing, implemented in commercial platforms such as D-Wave [1], has shown particular effectiveness in obtaining approximate solutions for large-scale combinatorial optimization tasks.

In the context of RDBMS, quantum annealing was first employed for *Multi-Query Optimization (MQO)* [40]. In this approach, the MQO instance is reformulated into a mathematical structure compatible with the constrained input format required by quantum

annealers and subsequently executed on a D-Wave processor [1]. The initial experiments demonstrated that, for certain query workloads, the quantum annealer achieved performance up to three orders of magnitude faster than classical optimization techniques. This framework was later extended to other database optimization problems, including transaction scheduling [5] and join-order optimization [36]. Specifically, in [36], a detailed evaluation was conducted across multiple query graphs for three-relation chain queries using the D-Wave platform. The study found that approximately 41% of the generated join orders were valid, with 10% of those corresponding to the optimal plans.

A more versatile and extensively studied paradigm in quantum optimization is based on *variational algorithms*, such as the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE). These algorithms employ a hybrid quantum-classical workflow in which the quantum processor evaluates a cost function, while a classical optimizer iteratively tunes the parameters of the quantum circuit to minimize it. The variational approach has been successfully employed in *join-order optimization* [25, 36, 37] and *index configuration selection* [22, 41]. In both applications, the underlying optimization problem is formulated as a Quadratic Unconstrained Binary Optimization (QUBO) model and solved using QAOA, yielding significant improvements in solution quality compared to classical heuristics.

An alternative strategy for *index selection* is proposed in [22], where the problem is formulated as a fully enumerative search across the exponential configuration space and solved using *Grover's Search*. The central innovation lies in the design of a quantum oracle capable of performing computations on classical data encoded within the *relative phase* of a quantum superposition state. This unconventional formulation enables intrinsic computation of index configuration parameters and facilitates efficient identification of qualifying configurations.

Recently, these ideas have been extended toward a more comprehensive quantum treatment of index selection. Advancing beyond prior quantum approaches that focused only on configuration selection, the *enumeration* and *costing* stages of the advisor pipeline have also been migrated into the quantum framework. Experimental results show that the proposed method recommends substantially better index configurations, outperforming commercial tools by

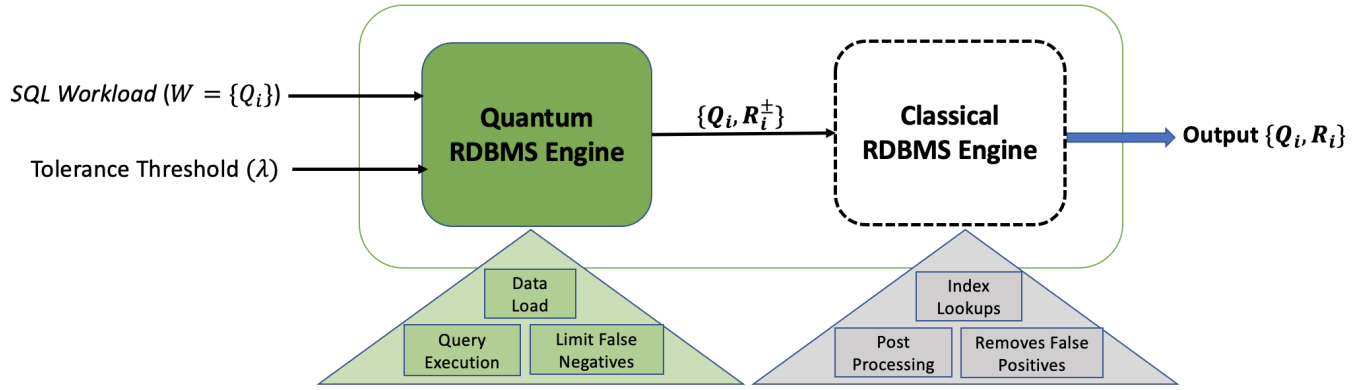


Figure 3: Quantum-Classical Hybrid SQL Query Contract

3 – 4× and previous quantum solutions by 2 – 3×, while producing configurations that are also cheaper to construct. These outcomes underscore the growing promise of quantum computing in enhancing the efficiency and quality of physical database design.

The above alternative approaches are an exemplar of how a given problem can be solved using different quantum properties, showcasing the versatility and potential of quantum computing in solving complex database management challenges.

2.4 Stage IV: Quantum Query Execution

In this segment, we discuss proposals for executing SQL queries on quantum computers. The earliest concepts were introduced in [8], which outlined an abstract framework for implementing basic relational operations, such as selection, projection, and join using quantum primitives. Subsequently, [45] proposed a set of *Quantum Query Language (QQL)* operations encompassing Insert, Update, Delete, Select, as well as database backup and recovery. However, this model was limited to an implicit database containing a single column of unique values. Later, [21] extended this line of work by proposing a mechanism to load multi-column relational tables and employ Grover’s Search to identify qualifying tuples. The approach was further expanded in [15] to perform inner joins between relational tables, again using Grover’s Search to locate the tuples satisfying the join conditions.

We review the aforementioned literature and analyze the challenges that must be addressed to make quantum query execution a practical reality. We also examine a *weakened SQL query execution contract* proposed in [23], which relaxes the strict correctness requirement by introducing a small, non-zero tolerance threshold (λ) for result accuracy. This allows limited deviations in query outputs, thereby accommodating the probabilistic nature of quantum computation. The proposed *quantum-classical hybrid RDBMS contract* is shown in Figure 3.

In this model, the quantum engine produces intermediate results denoted as $\{Q_i, R_i^\pm\}$, where $R_i^\pm$ represent the presence of false positives and false negatives that may arise during quantum execution. These intermediate outputs are subsequently processed by

the classical RDBMS engine, which applies post-processing techniques, such as index lookups and validation filters, to minimize inaccuracies and produce the final output $\{Q_i, R_i\}$.

Finally, we highlight a complementary line of research focusing on the design of *quantum-inspired* algorithms for RDBMS [27, 34, 38]. These algorithms execute on classical hardware but leverage concepts from quantum computing to enhance performance and scalability. For example, [34] explores resource allocation in traditional relational databases within an OLTP environment, drawing inspiration from quantum superposition and measurement to enable resource transactions to commit without explicitly binding to concrete resource instances. This behavior is achieved by maintaining an intrinsic representation of all feasible “possible worlds,” each corresponding to a valid concrete resource assignment.

Another recent work has proposed a quantum-inspired approach that scales MQO to industrially relevant sizes [38]. The framework combines classical computation with specialized annealing hardware such as Fujitsu’s Digital Annealer [3], partitioning large MQO instances into manageable sub-problems while preserving global dependencies through a Dynamic Search Steering (DSS) mechanism. Empirical evaluations demonstrate that this incremental annealing strategy achieves substantial runtime reductions and solution quality improvements, solving workloads with up to a thousand queries and outperforming both heuristic baselines and conventional quantum annealing approaches. These results highlight the practicality of hybrid quantum-inspired optimization for large-scale database management problems.

2.5 Stage V: Quantum Advisor Demonstration

In this segment, we demonstrate the inner workings of a recently proposed Quantum Index Advisor [22] through an interactive demonstration designed for learning purposes. The demonstration allows the audience to visually observe how the system processes an input workload and constructs the corresponding quantum circuit step by step. At each stage of circuit construction, the demo performs mid-circuit evaluation to display the intermediate quantum state and illustrates how to interpret these results.

This session is intended to help quantum newcomers understand how quantum circuits encode and manipulate information within

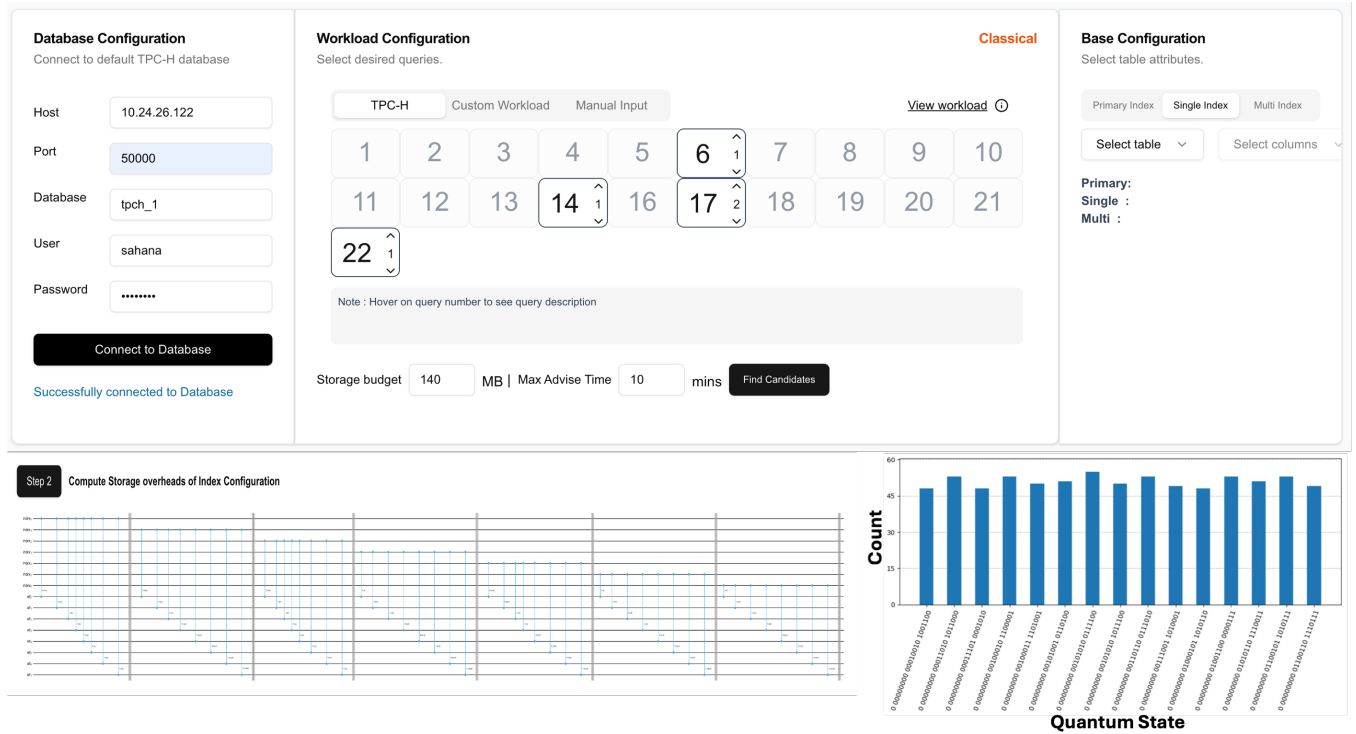


Figure 4: Demonstration of Quantum Index Advisor

a superposition state. By visualizing the transformation of quantum states and the role of individual gates, the audience gains an intuitive grasp of how quantum principles can be applied to database tuning tasks. Figure 4 presents sample snapshots from the demonstration. We expect that this exercise will familiarize participants with the key ideas behind quantum database algorithms and prepare them for the concluding hands-on session.

2.6 Stage VI: Hands-On Quantum Exercise

In this final segment, we move from theoretical concepts to a hands-on exercise in constructing practical applications, using a guided Jupyter notebook. As an exemplar, we implement the two Index Selection quantum algorithms proposed in [22], one based on Grover Search, and the other on QAOA. This interactive session offers a unique opportunity for participants to code and solve an instance of an RDBMS problem using two completely different quantum algorithms, gaining first-hand experience of distinct quantum approaches to a common challenge.

We will guide the participants through the process of setting up the quantum environment, designing quantum circuits, and executing these circuits on the **Qiskit simulator** [16] to understand their behavior and results. Following the simulator experience, the same algorithms will then be executed on real quantum hardware – specifically, the **133-qubit IBM Heron Circuit processor** [17]. By engaging with these practical scenarios, participants gain direct experience in quantum database programming, understanding the nuances of quantum circuit design, and its performance evaluation.

2.7 Target Learning Outcomes

Our expectation is that at the conclusion of this tutorial, participants will have a good grounding in both conceptual understanding and application of quantum computing principles to database management problems. Specifically, they will:

- Understand the fundamental building blocks of quantum computing, including qubits, superposition, entanglement, and quantum measurement, and how these differ from classical computing concepts.
- Recognize the architectural and algorithmic challenges involved in adapting core DBMS components, such as query optimization, physical design, and transaction scheduling, to quantum and quantum-inspired settings.
- Learn how optimization problems in database systems (e.g., join-order optimization, multi-query optimization, and index configuration selection) can be solved using quantum algorithms such as QAOA or Grover Search.
- Explore end-to-end implementations of quantum database algorithms through guided demonstrations, including mid-circuit visualization and state interpretation.
- Develop practical familiarity with constructing and executing quantum circuits on both simulators and real quantum hardware using Jupyter notebook-based workflows.
- Gain insight into current research frontiers, including hybrid quantum-classical approaches and quantum-inspired optimization, and become aware of key open challenges in realizing quantum-native database engines.

Collectively, these outcomes aim to demonstrate how traditional database system problems can be reformulated and solved using emerging quantum computing paradigms, equipping participants with the conceptual understanding and technical foundation needed to contribute to this rapidly evolving field.

3 Target Audience

Hosting relational database engines on quantum platforms is a new and exciting multidisciplinary area that lies at the intersection of RDBMS and quantum computing. This tutorial is expected to attract a broad audience at SIGMOD 2026, including researchers, developers, and students interested in the internals of database engines and the emerging applications of quantum computing. The intended background corresponds to that of an introductory database systems course, covering relational data models and fundamental query optimization and processing techniques. While prior exposure to quantum computing is advantageous, the tutorial will introduce the essential concepts necessary to understand the interaction between quantum computation and data management.

Database researchers can expect this tutorial to provide fresh perspectives on a long-standing research area, potentially motivating new directions for building more efficient database engines. For quantum computing enthusiasts, the tutorial highlights an important and emerging application domain of quantum technologies. System developers and practitioners will find the presented techniques valuable for rethinking and redesigning system architectures. Finally, database instructors and students will benefit from understanding and appreciating the fusion of database systems and quantum computing, extending beyond the usual theoretical discussions in classroom settings.

We hope this tutorial will motivate new Ph.D. students and researchers to pursue ambitious research directions beyond conventional middleware topics. The tutorial materials comprise the papers discussed in the presentation, complemented by *Qiskit SQK* [19] and official *Qiskit tutorials* [2], which serve as an introduction to fundamentals of quantum computing. The tutorial aims to bridge the gap between theory and practice by demonstrating how quantum primitives can be leveraged to rethink core DBMS components.

4 Presenter Details

Manish Kesarwani is an Advisory Research Scientist at IBM Research India. With over a decade of research experience at IBM Research, he has led projects at the intersection of RDBMS, Applied Cryptography, AI, and Quantum Computing. He holds 45+ patents, has served on program and organizing committees of top-tier conferences, and was recognized among IBM's Top Global Contributors and Top Inventors in 2025. His recent work focuses on fundamental challenges in hosting RDBMS components on quantum platforms, some of which have appeared in premier DBMS conferences.

Jayant Haritsa is on the faculty of the Department of Computational & Data Sciences and the Department of Computer Science & Automation at the Indian Institute of Science, Bangalore, since 1993. He received his undergraduate degree from IIT Madras, and the PhD degree from U. Wisconsin – Madison. He is a Fellow of ACM and IEEE for his contributions to the design and implementation of database engines. He was the Program Co-Chair for VLDB 2016 and ICDE 2010, and is now General Co-Chair for SIGMOD 2026.

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